

TrES-1b

R-1232

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-1_210509 · created 2026-07-31T09:58:54+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2459343.9126 +/- 0.0024 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.127 +/- 0.013
Transit depth (Rp/Rs)^2	1.62 +/- 0.34 [%]
Orbital Inclination (inc)	89.82 +/- 0.86
Airmass coefficient 1 (a1)	1.325 +/- 0.014
Airmass coefficient 2 (a2)	-0.221 +/- 0.015
Scatter in the residuals of the lightcurve fit is	1.07 %
Best Comparison Star	None
Optimal Aperture	4.03
Optimal Annulus	13.91
Transit Duration (day)	0.1025 +/- 0.002

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.127 ± 0.013 vs 0.1358 (deviation 0.68σ)
Duration fit vs published	0.1025 d vs 0.1042 d (deviation 0.83σ)
Mid-time O–C	5.1 min
Depth SNR	9.8
Residual scatter	1.07 %

Light curve

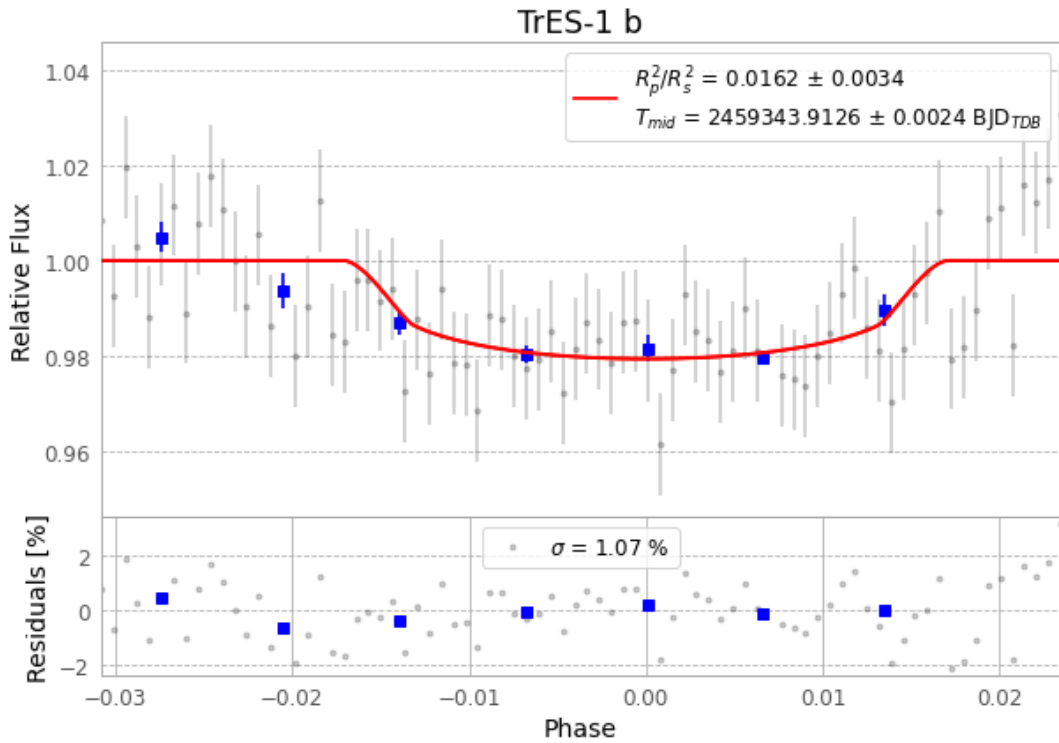


Figure 1 — FinalLightCurve_TrES-1 b_2021-05-09.png (SHA-256 f80649b7538889fc...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [349, 220], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 11e68f91e3328eed...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

80 FITS frames (53 MB), TRES-1210509073612.FITS → TRES-1210509113312.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-1-210509.log (34bdf1627f87...), FinalLightCurve_TrES-1 b_2021-05-09.png (f80649b75388...), FinalLightCurve_TrES-1 b_2021-05-09.csv (f81dd388b3ab...)

Compute resources

Wall time 135.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-31 09:58Z.